

2.4 Propositional Logic

Propositional logic is a basic form of logic used in computer science. It deals with statements that can be true or false. These statements are called **propositions**.

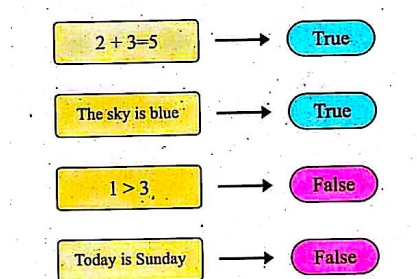


Figure 2.3: Propositional Logic Statements

DO YOU KNOW? Propositional logic is the simplest type of logic, but it forms the base of all computer reasoning!

Proposition

A proposition is a statement that is either true or false. It cannot be both true and false at the same time.

- Questions and commands are **not** propositions.
- Example: "The sun rises in the east" is a proposition.
- A proposition must have a clear meaning and should not be confusing.
- Propositions are the basic units of logic.
- A proposition is usually represented by a capital letter such as P, Q, or R.
Example: P = "The sky is blue"

Simple and Compound Propositions

- **Simple Proposition:** Contains only one statement. It cannot be broken into smaller parts. Example: "It is raining"
- **Compound Proposition:** Formed by combining two or more simple propositions using logical connectives. Example: "It is raining and it is cold"

Compound propositions can express complex ideas and are common in logical reasoning.

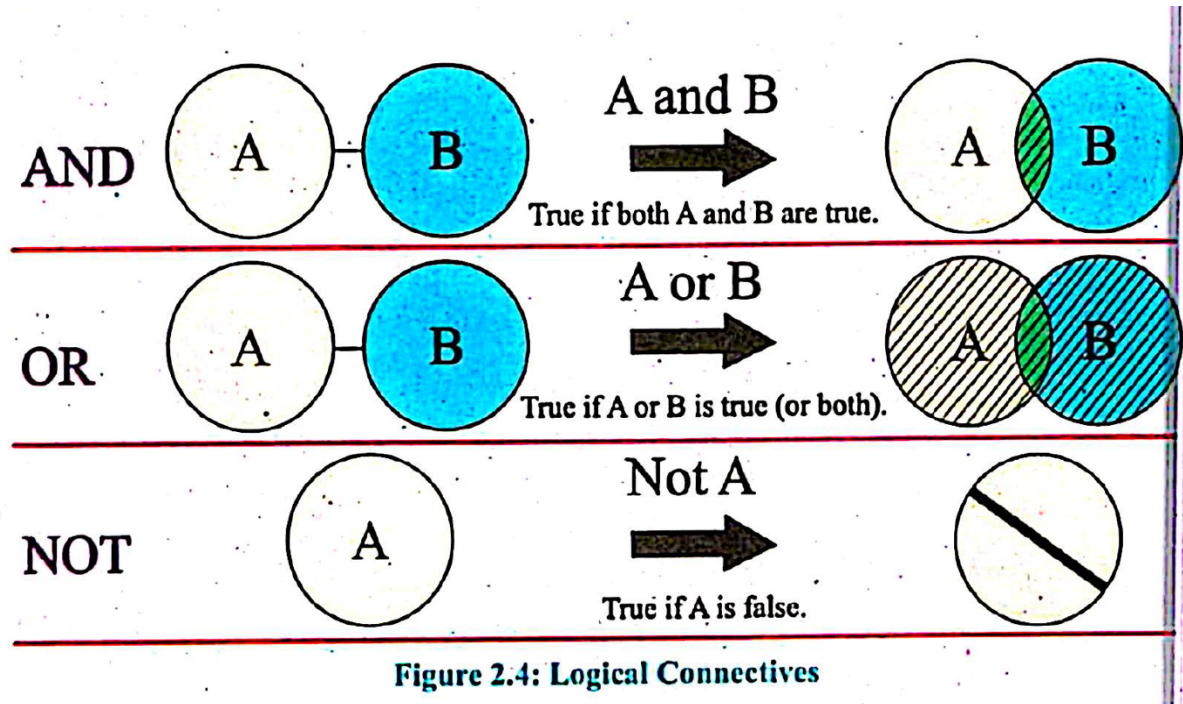
Logical Connectives: AND, OR, NOT

Logical connectives are words used to connect propositions.

Connective	Meaning	Result Condition
AND ($\wedge$)	Both statements must be true	True only if both A and B are true
OR ($\vee$)	At least one statement is true	True if A or B (or both) is true

NOT ($\neg$)	Reverses the truth of a statement	True if A is false
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These connectives help form compound propositions and are commonly used in computer programs for decision-making.



Short Questions

1. What is propositional logic?
2. What is a proposition?
3. Give two examples of propositions.
4. Why are questions and commands not considered propositions?
5. Differentiate between simple and compound propositions.
6. What are logical connectives? Name them.
7. Explain the AND connective with an example.
8. Explain the OR connective with an example.
9. Explain the NOT connective with an example.
10. How are propositions usually represented?